Chapter 4
Hierarchical Models

Xavi Puig Oriol
Departament d'Estadística i I.O.
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Chapter 4. Hierarchical Models

Bayesian Model (non-hierarchical)

$$p(\tilde{y} \mid \theta)$$
$$\pi(\theta)$$

Hierarchical Bayesian Model

$$p(\tilde{y} \mid \theta) \longleftarrow \text{1st Level}$$
$$\pi(\theta \mid \gamma) \longleftarrow \text{2nd Level}$$
$$\pi(\gamma) \longleftarrow \text{3rd Level}$$

The hierarchical Bayesian Model treats the parameters of the prior distribution as random variables. This fact adds a new level to the model.

Chapter 4. Hierarchical Models

If we take a random sample of size n, $y_1, \ldots, y_n$, then we can write the hierarchical bayesian model as:

$$\tilde{y}_i \mid \theta_i \sim p(\tilde{y}_i \mid \theta_i) \qquad \textit{for } i=1,..,n$$
$$\theta_i \mid \gamma \sim \pi(\theta_i \mid \gamma)$$
$$\gamma \sim \psi(\gamma)$$

$\tilde{y}_i$	observable variable	$p(\tilde{y}_i \mid \theta_i)$	probability distribution
θ_i	parameter	$\pi(\theta_i \mid \gamma)$	prior distribution
γ	hyperparameter	$\psi(\gamma)$	hyperprior distribution

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$$\theta_i \mid \gamma \sim \pi(\theta_i \mid \gamma)$$
$$\gamma \sim \psi(\gamma)$$

Or in an equivalent way as:

$$\tilde{y} = (\tilde{y}_1, \ldots, \tilde{y}_n) \mid \theta = (\theta_1, \ldots, \theta_n) \sim \prod_{i=1}^{n} p(\tilde{y}_i \mid \theta_i)$$
$$\theta = (\theta_1, \ldots, \theta_n) \mid \gamma \sim \prod_{i=1}^{n} \pi(\theta_i \mid \gamma)$$
$$\gamma \sim \psi(\gamma)$$

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$$\theta = (\theta_1, \ldots, \theta_n) \mid \gamma \sim \prod_{i=1}^{n} \pi(\theta_i \mid \gamma)$$

$$\gamma \sim \psi(\gamma)$$

- The y 's are conditionally independent but not identically distributed.
- The θ 's are conditionally independent and identically distributed

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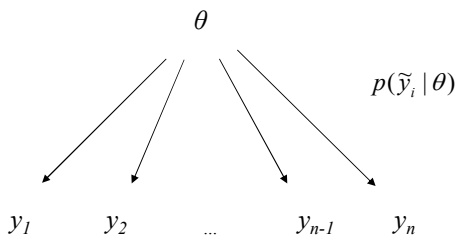
Bayesian Model (non-hierarchical)

$$\tilde{y}_i \mid \theta \sim p(\tilde{y}_i \mid \theta)$$

$$\theta \sim \pi(\theta)$$



The y 's are generated by the probability model $p(\tilde{y}_i \mid \theta)$

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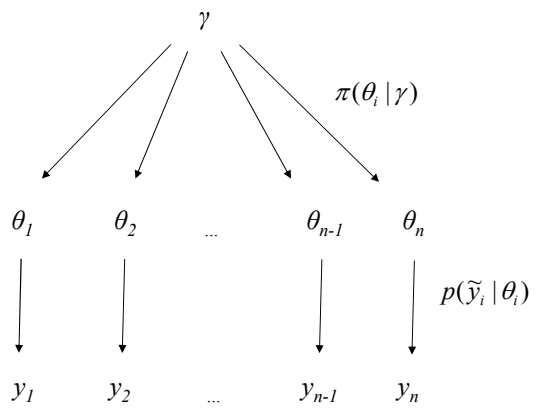
Hierarchical Bayesian Model

$$\tilde{y}_i \mid \theta_i \sim p(\tilde{y}_i \mid \theta_i)$$
$$\theta_i \mid \gamma \sim \pi(\theta_i \mid \gamma)$$
$$\gamma \sim \psi(\gamma)$$



The y's are generated by the probability model $p(\tilde{y}_i \mid \theta_i)$, and the θ's are generated by $\pi(\theta_i \mid \gamma)$

Chapter 4. Hierarchical Models

Example 1: Hospitals

Exercise 6.2 Surgical: Institutional ranking. This exercise considers mortality rates in 12 hospitals performing cardiac surgery in babies. The data are shown below:

Hospital	No of ops	No of deaths
A	47	0
B	148	18
C	119	8
D	810	46
E	211	8
F	196	13
G	148	9
H	215	31
I	207	14
J	97	8
K	256	29
L	360	24

The objective of this study is to know the probability of death around all the hospitals in the country, not only in the hospitals that are in the sample.

Chapter 4. Hierarchical Models

Example 1: Hospitals

$y_i :=$ number of deaths in the *i-th* hospital
$n_i :=$ number of surgeries in the *i-th* hospital

for i=1,..,12

Model A

$$\tilde{y}_i \mid \theta \sim Binomial(n_i, \theta)$$
$$\theta \sim Beta(a = 1, b = 1)$$

Chapter 4. Hierarchical Models

Example 1: Hospitals

$y_i :=$ number of deaths in the *i-th* hospital
$n_i :=$ number of surgeries in the *i-th* hospital

for i=1,..,12

Model A

$$\tilde{y}_i \mid \theta \sim Binomial(n_i, \theta)$$
$$\theta \sim Beta(a = 1, b = 1)$$

Model A'

$$\sum \tilde{y}_i \mid \theta \sim Binomial(\sum n_i, \theta)$$
$$\theta \sim Beta(a = 1, b = 1)$$

Are models A and A' equivalent?

Chapter 4. Hierarchical Models

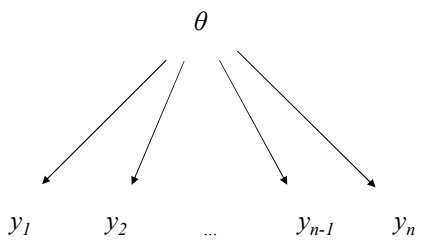
Example 1: Hospitals

$y_i :=$ number of deaths in the *i-th* hospital
$n_i :=$ number of surgeries in the *i-th* hospital

for i=1,..,12

Model A

$$\tilde{y}_i \mid \theta \sim Binomial(n_i, \theta)$$
$$\theta \sim Beta(a = 1, b = 1)$$



Where θ is the probability of dying, and it is the same for all hospitals

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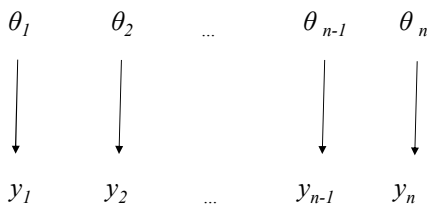
Example 1: Hospitals

Model B

$$\tilde{y}_i \mid \theta_i \sim Binomial(n_i, \theta_i)$$
$$\theta_i \sim Beta(1, 1)$$



Where θ_i is the probability of dying for the *i-th* hospital for i=1,...,12, but we don't know anything about the death probability from other hospitals.

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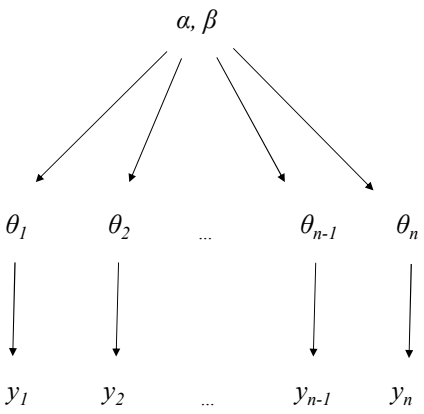
Example 1: Hospitals

Model C

$$\tilde{y}_i \mid \theta_i \sim Binomial(n_i, \theta_i)$$
$$\theta_i \mid \alpha, \beta \sim Beta(\alpha, \beta)$$
$$\alpha \sim Gamma(0.01, 0.001)$$
$$\beta \sim Gamma(0.01, 0.001)$$



We can make inference and prediction at different levels

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Example 1: Hospitals

Model C (hierarchical):

All we know about the probability of dying in the *i-th* hospital is in :

$$\pi(\theta_i \mid y)$$

And we can make inference of the probability of dying in a hospitat that is not in the sample using the posterior predictive distribution of the second level:

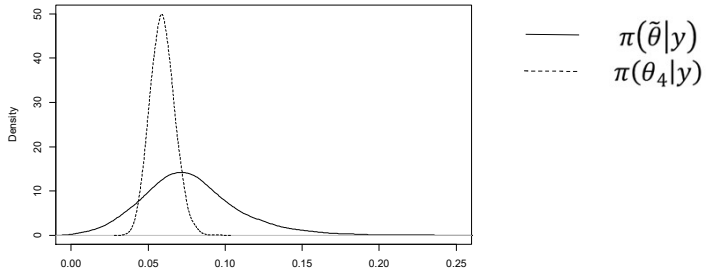
$$\pi(\tilde{\theta} \mid y) = \iint \pi(\tilde{\theta} \mid \alpha, \beta)\pi(\alpha, \beta \mid y)\partial\alpha\partial\beta$$

Chapter 4. Hierarchical Models

Example 1: Hospitals

Posterior probability distribution of the probability of dying for hospital n° 4

Posterior probability distribution of the probability of dying for a new hosptial



Observation:

$$E(\theta_i|y) = \lambda_i \frac{y_i}{n_i} + (1-\lambda_i)E(\tilde{\theta}|y)$$ where λ_i increases when n_i increases

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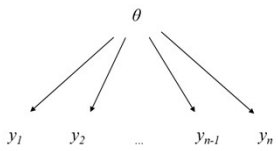
Example 1: Hospitals

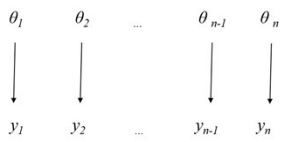
Model A

$$\tilde{y}_i \,|\, \theta_i \sim Binomial(n_i, \theta)$$
$$\theta \sim Beta(1,1)$$



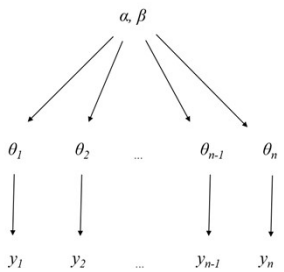
Model B

$$\tilde{y}_i \,|\, \theta_i \sim Binomial(n_i, \theta_i)$$
$$\theta_i \sim Beta(1,1)$$



Model C

$$\tilde{y}_i \,|\, \theta_i \sim Binomial(n_i, \theta_i)$$
$$\theta_i \,|\, \alpha, \beta \sim Beta(\alpha, \beta)$$
$$\alpha \sim Gamma(0.01, 0.001)$$
$$\beta \sim Gamma(0.01, 0.001)$$



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Example 1: Hospitals

Model A

$$\tilde{y}_i \mid \theta_i \sim Binomial(n_i, \theta)$$
$$\theta \sim Beta(1,1)$$

Model B

$$\tilde{y}_i \mid \theta_i \sim Binomial(n_i, \theta_i)$$
$$\theta_i \sim Beta(1,1)$$

Model C

$$\tilde{y}_i \mid \theta_i \sim Binomial(n_i, \theta_i)$$
$$\theta_i \mid \alpha, \beta \sim Beta(\alpha, \beta)$$
$$\alpha \sim Gamma(0.01, 0.001)$$
$$\beta \sim Gamma(0.01, 0.001)$$

parameters

$$\begin{array}{c|c|} & \theta \\ \text{simulations} & \begin{array}{c} \theta^{(1)} \\ \theta^{(2)} \\ \vdots \\ \theta^{(M)} \end{array} \end{array}$$

parameters

$$\begin{array}{c|cccc|} & \theta_1 & \theta_2 & \dots & \theta_{12} \\ \text{simulations} & \theta_1^{(1)} & \theta_2^{(1)} & \dots & \theta_{12}^{(1)} \\ & \theta_1^{(2)} & \theta_2^{(2)} & \dots & \theta_{12}^{(2)} \\ & \vdots & & & \\ & \theta_1^{(M)} & \theta_2^{(M)} & \dots & \theta_{12}^{(M)} \end{array}$$

parameters

$$\begin{array}{c|cccccc|} & \theta_1 & \theta_2 & \dots & \theta_{12} & \alpha & \beta \\ \text{simulations} & \theta_1^{(1)} & \theta_2^{(1)} & \dots & \theta_{12}^{(1)} & \alpha^{(1)} & \beta^{(1)} \\ & \theta_1^{(2)} & \theta_2^{(2)} & \dots & \theta_{12}^{(2)} & \alpha^{(2)} & \beta^{(2)} \\ & \vdots & & & & & \\ & \theta_1^{(M)} & \theta_2^{(M)} & \dots & \theta_{12}^{(M)} & \alpha^{(M)} & \beta^{(M)} \end{array}$$

Chapter 4. Hierarchical Models

Example 1: Hospitals

$$\tilde{y}_i \mid \theta_i \sim Binomial(n_i, \theta_i)$$
$$\theta_i \mid \alpha, \beta \sim Beta(\alpha, \beta)$$
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parameters

$$\begin{array}{c|cccccc|} & \theta_1 & \theta_2 & \dots & \theta_{12} & \alpha & \beta \\ \text{simulations} & \theta_1^{(1)} & \theta_2^{(1)} & \dots & \theta_{12}^{(1)} & \alpha^{(1)} & \beta^{(1)} \\ & \theta_1^{(2)} & \theta_2^{(2)} & \dots & \theta_{12}^{(2)} & \alpha^{(2)} & \beta^{(2)} \\ & \vdots & & & & & \\ & \theta_1^{(M)} & \theta_2^{(M)} & \dots & \theta_{12}^{(M)} & \alpha^{(M)} & \beta^{(M)} \end{array}$$

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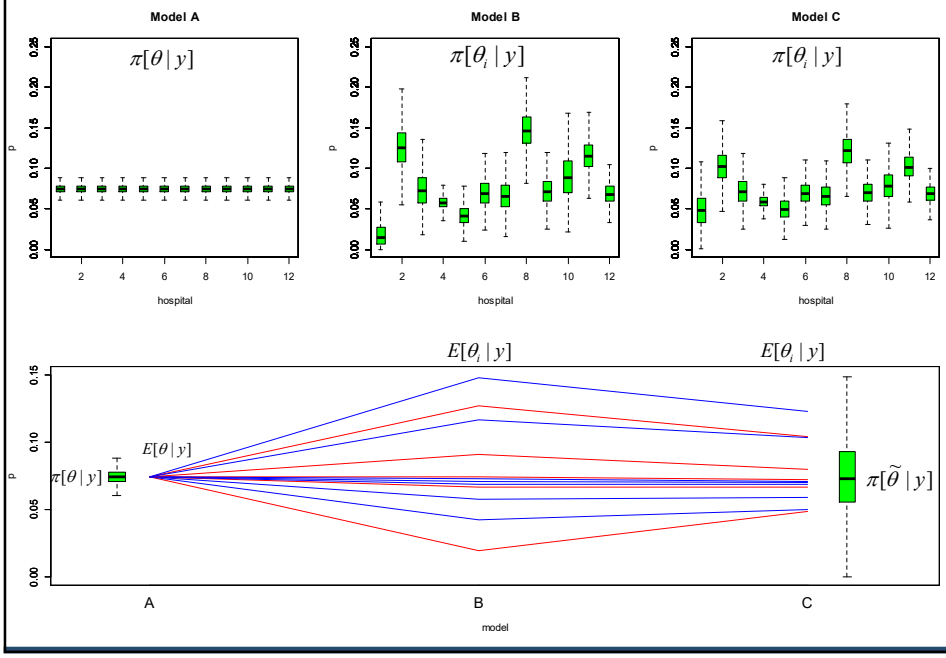


Chapter 4. Hierarchical Models

Observation: we can convert a hierarchical model into a non-hierarchical model, but we lose the option of making inference at the second level:

Hierarchical

$$\tilde{y} = (\tilde{y}_1, \ldots, \tilde{y}_n) \mid \theta = (\theta_1, \ldots, \theta_n) \sim \prod_{i=1}^{n} p(\tilde{y}_i \mid \theta_i)$$

$$\theta = (\theta_1, \ldots, \theta_n) \mid \gamma \sim \prod_{i=1}^{n} \pi(\theta_i \mid \gamma)$$

$$\gamma \sim \psi(\gamma)$$

Non-hierarchical

$$\tilde{y} = (\tilde{y}_1, \ldots, \tilde{y}_n) \mid \gamma \sim p(\tilde{y} \mid \gamma) = \prod_{i=1}^{n} \int_{\Omega_i} p(\tilde{y}_i \mid \theta_i) \pi(\theta_i \mid \gamma) \partial \theta_i$$

$$\gamma \sim \psi(\gamma)$$

Chapter 4. Hierarchical Models

Example 1: Hospitals

In the hospital example, we can rewrite the model at two levels as:

$$\begin{aligned} \tilde{y}_i \mid \theta_i &\sim Binomial(n_i, \theta_i) \\ \theta_i \mid \alpha, \beta &\sim Beta(\alpha, \beta) \\ \alpha &\sim Gamma(0.01, 0.001) \\ \beta &\sim Gamma(0.01, 0.001) \end{aligned} \quad \Rightarrow \quad \begin{aligned} \tilde{y}_i \mid \alpha, \beta &\sim BetaBinomial(n_i, \alpha, \beta) \\ \alpha &\sim Gamma(0.01, 0.001) \\ \beta &\sim Gamma(0.01, 0.001) \end{aligned}$$

The two-level model can make inference for the hyperparameters but cannot make inference for the probability of dying for a specific hospital in the sample.

Chapter 4. Hierarchical Models

Example 1: Hospitals

Can we do a *ranking* for the sample hospitals?

Chapter 4. Hierarchical Models

Example 1: Hospitals

parameter

$$\begin{array}{c} \\ \text{simulations} \end{array} \begin{array}{cccc} \theta_1 & \theta_2 & \dots & \theta_{12} \\ \hline \theta_1^{(1)} & \theta_2^{(1)} & \dots & \theta_{12}^{(1)} \\ \theta_1^{(2)} & \theta_2^{(2)} & \dots & \theta_{12}^{(2)} \\ \vdots & & & \\ \theta_1^{(M)} & \theta_2^{(M)} & \dots & \theta_{12}^{(M)} \end{array}$$

Ranking

$$\begin{array}{cccc} R_1 & R_2 & \dots & R_{12} \\ \hline R_1^{(1)} & R_2^{(1)} & \dots & R_{12}^{(1)} \\ R_1^{(2)} & R_2^{(2)} & \dots & R_{12}^{(2)} \\ \vdots & & & \\ R_1^{(M)} & R_2^{(M)} & \dots & R_{12}^{(M)} \end{array}$$

In each simulation we create a new *Ranking* variable for each hospital

For example, if in the first simulation after sorting the values of θ's we get:

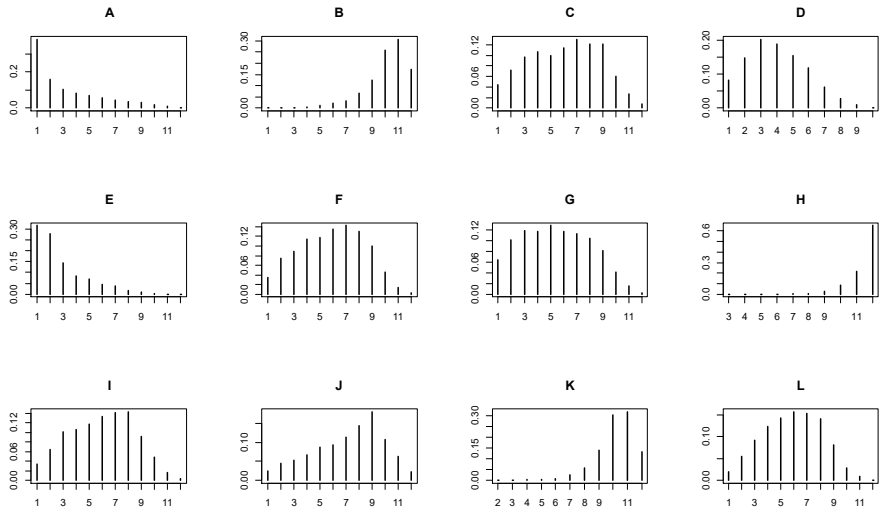
$$\theta_2^{(1)} < \theta_3^{(1)} < \theta_5^{(1)} < \theta_{12}^{(1)} < \theta_4^{(1)} < \theta_{11}^{(1)} < \theta_9^{(1)} < \dots < \theta_1^{(1)}$$

Then the ranking variable will be:

$$R_1^{(1)} = 12,\ R_2^{(1)} = 1, R_3^{(1)} = 2,\ R_4^{(1)} = 5, \dots, R_{12}^{(1)} = 4$$

Chapter 4. Hierarchical Models

The Posterior probability Distribution for the *ranking* variable for each hospital



80% CI

	A	B	C	D	E	F	G	H	I	J	K	L
10%	1	8	2	2	1	2	2	10	3	3	9	3
90%	7	12	9	7	6	9	9	12	9	10	12	9

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Example 1: Hospitals

Model C can be reparametrized as a Generalized Linear Model

Model C

$$\tilde{y}_i \mid \theta_i \sim Binomial(n_i, \theta_i)$$
$$\theta_i \mid \alpha, \beta \sim Beta(\alpha, \beta)$$
$$\alpha \sim Gamma(0.01, 0.001)$$
$$\beta \sim Gamma(0.01, 0.001)$$

Model C'

$$\tilde{y}_i \mid \theta_i \sim Binomial(n_i, \theta_i)$$
$$logit\left(\frac{\theta_i}{1-\theta_i}\right) = \beta_i$$
$$\beta_i \sim Normal(\mu, \sigma^2)$$
$$\mu \sim Normal(0, 10^{10})$$
$$\sigma^2 \sim InvGamma(0{,}001, 0{,}001)$$

This parameterization allows you to add explanatory variables to the model in a simple way.

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Example 2: sleepstudy

The data set record the reaction time of 18 subjects over 10 days period of sleep deprivation. This is a longitudinal study.

Part of the data:

Subject	Days	Time (ms)
1	0	249,6
1	1	258,7
1	2	250,8
1	3	321,4
1	4	356,9
1	5	414,7
1	6	382,2
1	7	290,1
1	8	430,6
1	9	466,4
2	0	205,3
2	1	203,3
...	...	...
18	9	364,1

The aim of the study is to find out how the reaction time changes when the days with sleep deprivation increase.

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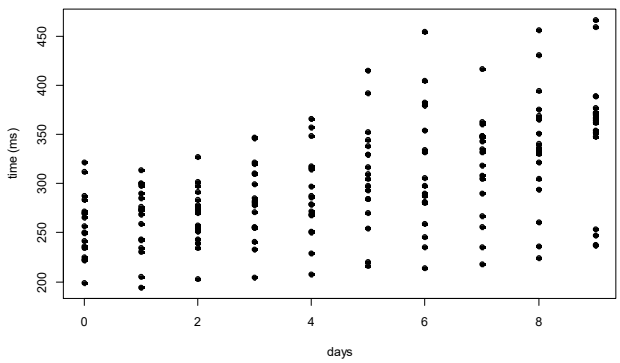
Example 2: sleepstudy

Reaction time over the days :



Chapter 4. Hierarchical Models

Example 2: sleepstudy

The response variable is continuous, so it seems reasonable to choose a normal model as statistical model. An initial statistical model could be:

$$y_{i,t}|\beta_0, \beta_1, \sigma, t \sim Normal(\beta_0 + \beta_1 t, \sigma)$$

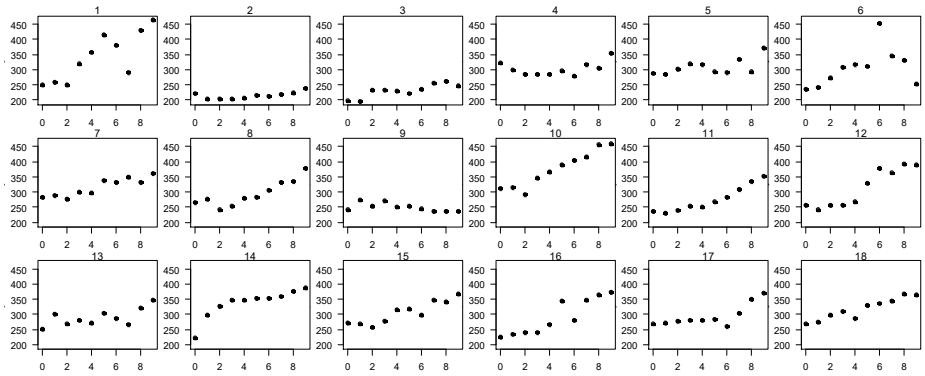
What do you think about the following questions?

- Could the basal reaction time among subjects be different?
- Could the relationship of reaction time and days be different among subjects?
- Are the observations independent?

Chapter 4. Hierarchical Models

Example 2: sleepstudy

Reaction time over the days in each subject:



Chapter 4. Hierarchical Models

Example 2: sleepstudy

This model takes into account that

- the basal reaction time among subjects could be different among subjects
- the relationship of reaction time and days be different among subjects

$$y_{i,t}|\beta_0, \alpha, \beta_1, \gamma, \sigma, \sigma_\alpha, \sigma_\gamma, t \sim Normal(\beta_0 + \alpha_i + (\beta_1 + \gamma_i)t, \sigma)$$

$$\alpha_i|\sigma_\alpha \sim Normal(0, \sigma_\alpha)$$

$$\gamma_i\,|\sigma_\gamma \sim Normal(0, \sigma_\gamma)$$

Chapter 4. Hierarchical Models

Example 2: sleepstudy

Defining the usual non-informative priors, we write the full Bayesian model as follows:

$$y_{i,t}|\beta_0, \alpha, \beta_1, \gamma, \sigma, \sigma_\alpha, \sigma_\gamma t \sim Normal(\beta_0 + \alpha_i + (\beta_1 + \gamma_i)t, \sigma)$$

$$\alpha_i|\sigma_\alpha \sim Normal(0, \sigma_\alpha)$$

$$\gamma_i\,|\sigma_\gamma \sim Normal(0, \sigma_\gamma)$$

$$\beta_0 \sim Normal(0, 1000)$$

$$\beta_1 \sim Normal(0, 1000)$$

$$\sigma^2 \sim InvGamma(0.001, 0.001)$$

$$\sigma_\alpha^2 \sim InvGamma(0.001, 0.001)$$

$$\sigma_\gamma^2 \sim InvGamma(0.001, 0.001)$$

Chapter 4. Hierarchical Models

I encourage you to read chapter 5 (Hierarchical models) and chapter 15 (*Hierarchical Lineal Models*) from the book:

Gelman A, Carlin J, Stern H, Dunson D, Vehtari A, and Rubin D (2020). Bayesian Data Analysis (3rd ed). London: Chapman & Hall.